

Open Source Hardware

In open source hardware, the hardware design, i.e., mechanical drawings, schematics, bills of material, PCB layout data, HDL source code and integrated circuit layout data are all released with the FOSS approach. Here are a few websites that will introduce you to some commonly-available open source hardware

arduino.cc

Arduino is an open source physical computing platform based on a simple microcontroller board. The Playground is a wiki where all the users of Arduino can contribute and benefit from their collective research. The Arduino official site has a section where you can find a list of Arduino-based DIY project ideas. The list is helpful for both the novice and the experienced. The project ideas have been divided into three categories for easy navigation, i.e., easy, intermediate and advanced.



<http://arduino.cc/>



beagleboard.org

The BeagleBoard is a low-power, low-cost, fan-less single-board computer. It is based on Texas Instruments' processors featuring the ARM Cortex-A series core. BeagleBoard.org is an all-volunteer activity started by a collection of passionate individuals, including several employees of Texas Instruments, interested in creating powerful, open and embedded devices.

The site has a discussion board along with links that provide information regarding where you can buy the board. It also refers to around 200 different projects related to the board. The BeagleBoard was designed with open source development in mind, and as a way of demonstrating the Texas Instruments' OMAP3530 system-on-a-chip. It is also sold to the public under the Creative Commons ShareAlike licence.

<http://beagleboard.org/>

oshwa.org

The Open Source Hardware Association aims to be the voice of the open hardware community, ensuring that technological knowledge is accessible to everyone. It encourages the collaborative development of technology that serves education, environmental sustainability and human welfare. The website educates individuals and the general public about open source hardware. It also encourages collaborative learning, knowledge exchange and social cohesion through details of conferences and other events focussed on open source hardware.



<http://www.oshwa.org/about/>



pandaboard.org

pandaboard.org is an online resource and collaboration site for the growing, active community supporting PandaBoard, the low-cost, open OMAP 4 processor-based mobile software development platform. The site is managed by volunteers who are committed to open source development. It provides developers opportunity to engage with community experts and source technical materials.

<http://www.pandaboard.org/>

olimex.com

OLinuXino is an open source software hardware, low-cost Linux-based, industrial-grade, single-board computer with GPIOs capable of operating at -25°C to +85°C. OLinuXino uses a widely-available microcontroller iMX233. This means that people can spin off their own boards economically and manufacture them easily as the processor in TQFP package is easy to assemble by a hobbyist. The project is hosted at Github and has an active forum for development discussion.



<https://www.olimex.com/Products/OLinuXino/open-source-hardware>